The quasar's bright light, seen by astronomers, comes from the disk's heated gas, Lynden-Bell concluded. Moreover, the magnetic field accelerates some of the gas's electrons to high energies; and the electrons then spiral around the magnetic force lines, emitting the quasar's observed radio waves.

Lynden-Bell worked out the details of all this using a combination of the Newtonian, relativistic, and quantum laws of physics. He easily explained everything about quasars that astronomers had seen, except their jets. His technical article describing his reasoning and his calculations (Lynden-Bell 1979) is one of the great astrophysics articles of all time.

The Jets: Extracting Power from Whirling Space

Over the next few years, astronomers discovered many more jets sticking out of quasars and studied them in great detail. It soon became clear that they are streams of hot, magnetized gas ejected from the quasar itself: from the black hole and its accretion disk (Figure 9.2). And the ejection is extremely powerful: the gas travels out the jets at nearly the speed of light. As it travels, and when it plows into material far from the quasar, the gas emits power in light, in radio waves, in X-rays, and even in gamma rays. The jets are sometimes as bright as the quasar itself, a hundred times brighter than the brightest galaxies.